

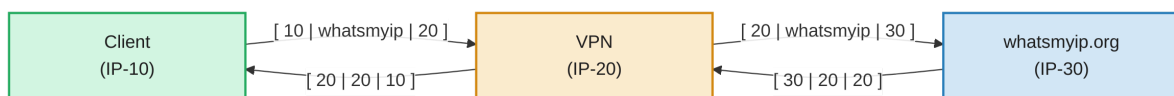
Lecture 6: Dark Web & Internet Working

VPNs route and secure traffic in a specific way, the Dark Web and Tor Browser rely on layered encryption for anonymity, and routers depend on **Subnet Masking** to decide paths efficiently.

VPN — Detailed Packet Flow

You can check your own IP address at services like whatismyip.org.

Example scenario: Client wants to reach whatismyip.org through a VPN.



What the ISP Actually Sees

A packet in transit generally looks like this to an observer:



- ✦ The **ISP knows the Source IP and Destination IP** — for example, it knows you connected to YouTube.
- ✦ However, it **does not know the content** — i.e., it knows you visited YouTube, but not which video you watched.

Is VPN Faster or Slower?

Generally, using a VPN makes the internet **slower**, because the VPN acts as an added **mediator** between the user and the destination:

```

Client → Server      (faster – direct path)
Client → VPN → Server (comparatively slower – extra hop)
  
```

Exception: When VPN Can Feel Faster

In some cases, an ISP deliberately **limits** certain traffic:

- ♦ **Streaming videos** — ISPs sometimes slow down streaming traffic specifically.
- ♦ **Restricted Wi-Fi (e.g. college networks)** — some websites are deliberately slowed down or blocked.

But a VPN changes your IP address, so:

- ♦ The ISP no longer knows *which website* you are visiting — it only sees that you're connecting to the **VPN**.
- ♦ Since the VPN itself isn't the thing being limited or blocked, your connection can end up feeling **faster** in these specific scenarios.

VPN Security — Not Always Fully Safe

A VPN adds an **extra security layer** around your data:

[data] → wrapped in an additional layer by the VPN

- ♦ The ISP **cannot read** the actual data, but it can sometimes **guess** things about it by observing the **pattern** of the data packets.
- ♦ The **VPN** itself can read your decrypted request — meaning a VPN is not automatically 100% safe. A dishonest or compromised VPN provider could disclose a user's **identity and browsing history**.

A truly privacy-respecting VPN does not keep logs of what users do. However, in some countries, VPNs and ISPs may be legally required to keep connection records for a period of time — so a "no logs" claim should always be read with that in mind.

Dark Web

*The **Dark Web** refers to parts of the internet that are not indexed by standard search engines and require special software (like Tor) to access.*

It is often associated with illegal activity, though it also contains legitimate content — access alone does not imply wrongdoing.

Tor Browser

- ♦ **Tor Browser** is used to access the Dark Web.
- ♦ Dark Web websites typically use the **.onion** domain extension.
- ♦ Tor was **originally developed by the US Military**, designed for **secure and anonymous** communication.

How Tor Achieves Anonymity — Onion Routing

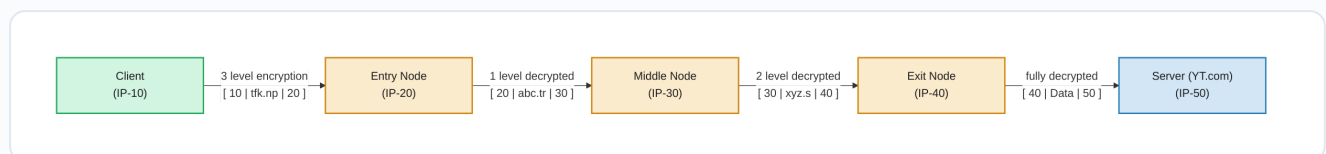
Tor routes traffic through multiple independent relay nodes, encrypting the data in **layers** (like an onion) so that no single node knows both who sent the message and what its final destination is.

3 Level Encryption

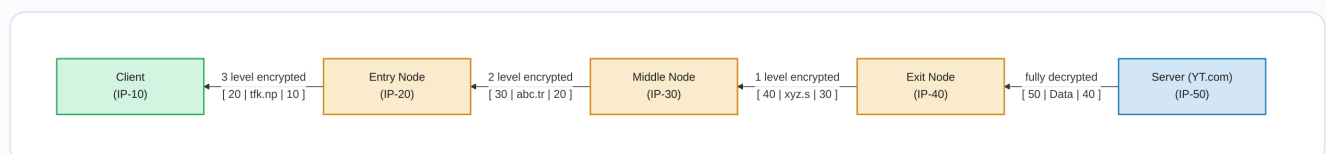
Before sending, the client wraps the request in three layers of encryption — each layer tells one node where to forward next, and only the innermost layer reveals the real destination and content:

```
Real destination: YT.com
Level 1 (innermost): xyz.s
Level 2: abc.tr
Level 3 (outermost): tfk.np
```

Request Flow (Client → Server)



Reply Flow (Server → Client)



*The **Client** has the authority to decrypt and view the final data, since it holds all the necessary decryption keys accumulated through the layered process.*

What Can the ISP See?

- ✦ The ISP knows that **client (10)** connected to **entry node (20)** — i.e., it knows you're using Tor.
- ✦ At most, it might glimpse a heavily encrypted fragment (e.g. "tfk.np") — but since it is **3-level (or more) encrypted**, it **cannot decode the actual destination or content**.
- ✦ Each Tor node is typically located in a **different country**, which makes it very difficult for any single observer (including an ISP) to trace the full path.

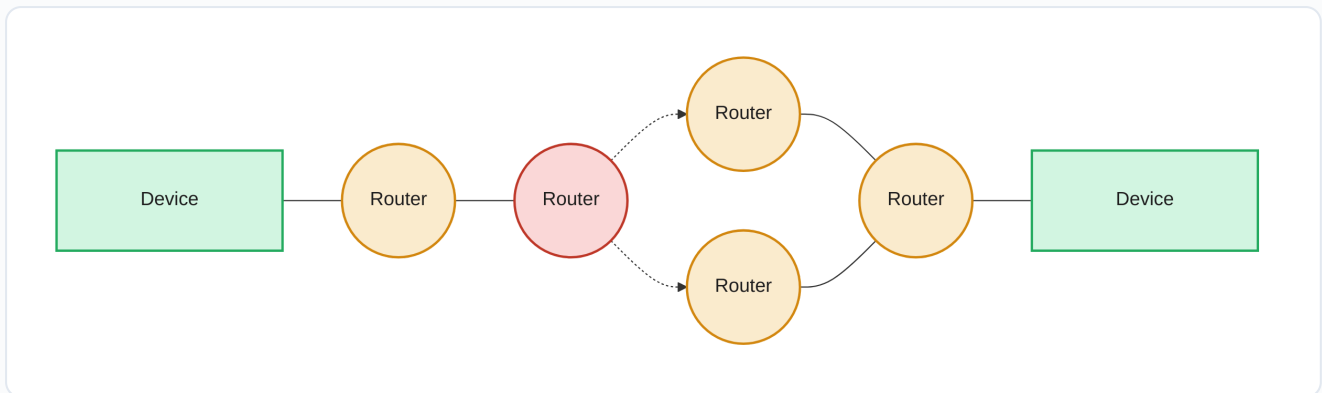
Important Notes on Tor

- ✦ **Tor** is not illegal in India — it also hosts many legitimate websites, not just illicit ones.
- ✦ The **Tor Browser** itself does not track your browsing history.
- ✦ If a device is under investigation, the underlying operating system might reveal some activity, but the **Tor Browser** does not maintain persistent logs — it refreshes and deletes its internal data roughly every **10 minutes**.
- ✦ Combining a **VPN** with **Tor Browser** provides double security — stronger than using either alone.

- ♦ While Tor typically uses a **3-node** architecture, some configurations use **6 nodes** for extra security — though this makes the connection very slow.

Efficient Routing — Why Random IPs Are a Problem

If IP addresses were assigned **completely randomly**, routers would find it very slow and inefficient to determine the correct path for every packet — because there would be no pattern to rely on.



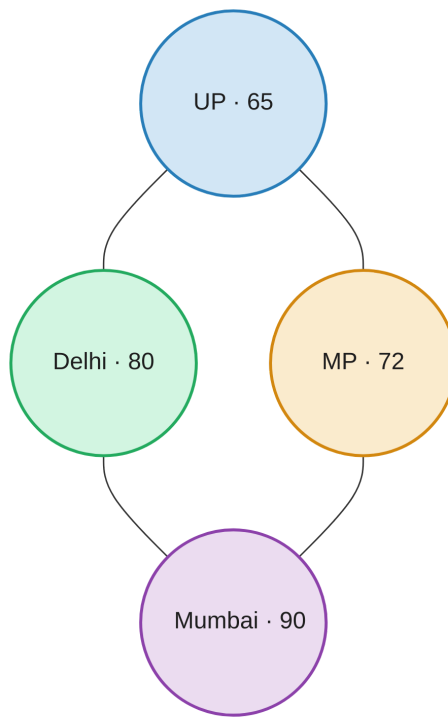
With no pattern to rely on, the router has no shortcut — it must check the destination address against a scattered, unordered list of devices one at a time until it finds a match.

Solution: Group Addresses by Region

By keeping the **same Network ID for one geographic region**, routers can make routing decisions **much faster** — since all addresses in a region share a recognizable prefix.

Example (illustrative):

UP	→ Network ID 65
Delhi	→ Network ID 80
MP	→ Network ID 72
Mumbai	→ Network ID 90



Grouping IP ranges by region makes packets "easily deliverable" — a router can quickly tell which broad direction to forward a packet just by checking its Network ID, without needing to know the exact final device.

Subnet Masking

Subnet Masking is the technique of dividing a single large network block into smaller **sub-networks**, so that IP address blocks can be distributed more precisely — for example, region by region.

Why It's Needed

Take the Class A block allocated to "Idea": 73.0.0.0

Host IDs available = $2^{24} - 2 \approx 1.67$ crore addresses

Giving this **entire** block to a single city (e.g. Delhi) wouldn't make sense — a single city may not have that many devices needing addresses. The solution is to **subnet** this block into smaller pieces.

Example: Splitting a Class C Block

Take a Class C network: 194.268.9.0

Total Host IDs (before subnetting) = $2^8 - 2 = 254$ (treated as up to 256 for simplicity in splitting)

Split into 2 equal halves (128 addresses each):

```
Block 1 → 194.268.9.0    - 194.268.9.127    (assigned to Delhi)
Block 2 → 194.268.9.128 - 194.268.9.255    (assigned to UP)
```

This is written using **subnet mask notation**:

194.268.9.0/25 → 194.268.9.127/25 (Delhi)
194.268.9.128/25 → 194.268.9.255/25 (UP)

What Does "/25" Mean?

The number after the slash indicates **how many bits are used for the Network ID**.

194 . 268 . 9 . 1
8bit 8bit 8bit 1bit = 25 bits total → Network ID
32 - 25 = 7 bits remaining → Host ID

- ✦ With **25 bits fixed** as Network ID, only the **last 7 bits** vary — giving $2^7 = 128$ possible addresses per subnet, matching our split above.

How the Range Is Derived

194.268.9.0		→ if remaining bits filled with 0 → 194.268.9.0
		→ if remaining bits filled with 1 → 194.268.9.127
194.268.9.1		→ if remaining bits filled with 0 → 194.268.9.128
		→ if remaining bits filled with 1 → 194.268.9.255

Another Example

220.68.90.0 → Network ID uses 24 bits, Host ID uses remaining 8 bits

Splitting further using one more bit:

```
220.68.90.0 ..... → if remaining bits filled with 0 → 220.68.90.0
                    → if remaining bits filled with 1 → 220.68.90.127
```

```
220.68.90.1 ..... → if remaining bits filled with 0 → 220.68.90.128
                  → if remaining bits filled with 1 → 220.68.90.255
```

Key Takeaway on Subnetting

- ✦ **Regions are divided using Subnet Masking**, allowing large address blocks to be broken into smaller, regionally meaningful chunks.
- ✦ **Routers store subnet mask information**, which lets them quickly determine which region (and eventually which device) a packet should be routed toward — making routing decisions fast and efficient, even with billions of devices online.

Summary

- ♦ A VPN routes traffic through an intermediate server, encrypting the request and changing the visible source IP; the ISP can still see the source and destination IPs (that you connected to a site) but not the actual content exchanged.
- ♦ VPNs are usually slower than a direct connection because they add an extra hop (mediator), but can feel faster in cases where the ISP specifically throttles or blocks certain traffic that the VPN bypasses.
- ♦ VPNs are not automatically fully safe — while the ISP can't read the data, the VPN provider itself can, so a VPN that keeps logs could potentially expose your identity and browsing history.
- ♦ The Dark Web refers to parts of the internet accessed via special software like the Tor Browser (.onion sites), originally developed by the US Military for secure, anonymous communication.
- ♦ Tor achieves anonymity through onion routing — traffic passes through an entry, middle, and exit node, with a layer of encryption removed at each hop, so no single node can see both the original sender and the final destination.
- ♦ Because Tor nodes are located in different countries and traffic is multiply encrypted, ISPs can typically tell that you're using Tor but cannot see what you accessed or decode the encrypted content.
- ♦ Tor is not illegal in India, doesn't track browsing history, and clears its internal data roughly every 10 minutes; combining VPN with Tor provides an extra layer of security.
- ♦ Randomly assigned IP addresses would make routing very slow; grouping addresses by region (same Network ID) lets routers make fast routing decisions.
- ♦ Subnet Masking divides a large network block into smaller sub-networks — written as a "/n" notation indicating how many bits form the Network ID, with the remaining bits used for Host ID within each subnet.
- ♦ Routers store subnet mask information, enabling them to quickly identify the right region/direction for a packet without needing to track every individual device address.